

Security, costs, and ownership

How the FASTR platform protects a country's data — and what it takes to run it

»» FASTR Useful Data

Data security

The image features a solid teal background. In the center, the words "Data security" are written in a bold, white, sans-serif font. To the right of the text, there is a large, faint, light-teal graphic consisting of several concentric circles. On the left side, there are two overlapping semi-circular shapes in a darker shade of teal. The overall design is modern and minimalist.

What the platform holds

- **Monthly service totals per facility** — for example, 45 first antenatal visits at one clinic in March
- The **same figures as the DHIS2 reports** the ministry already produces
- **No patient records** — no names, no addresses, nothing individual

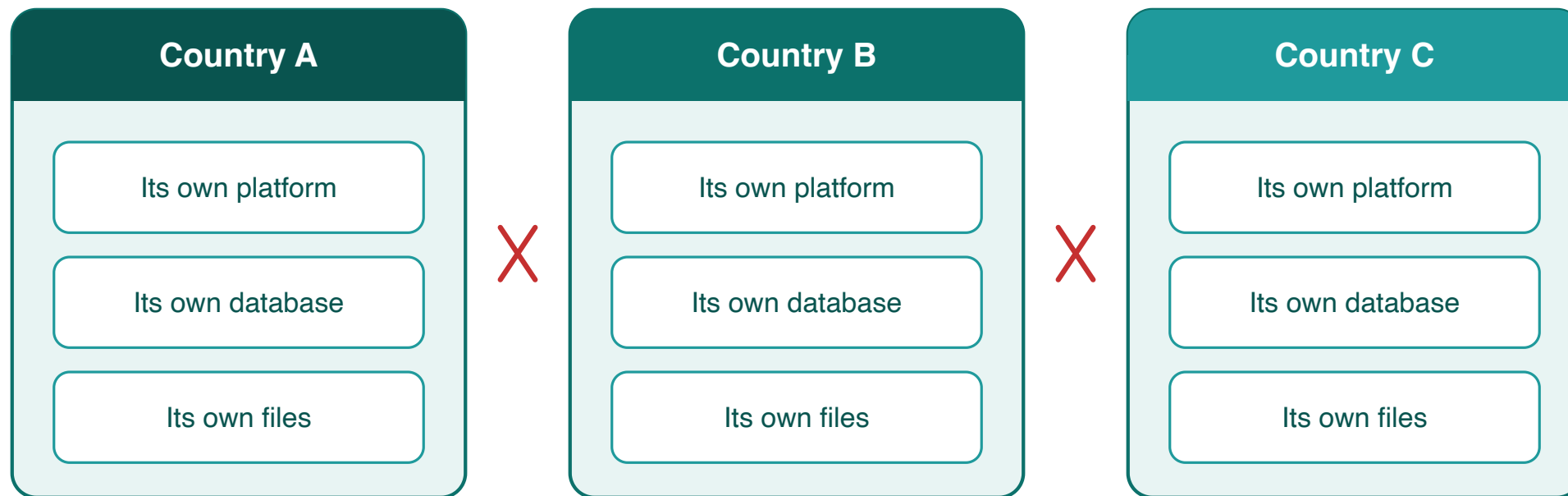
Four questions any government should ask

- Could our data leave the country's control?
- Could the wrong person get in?
- Could data be intercepted on the way — or lost in a failure?
- Could the AI expose what it sees?

The next four slides take these one at a time: the risk, then what answers it.

Could our data leave the country's control?

The risk: data flowing into a shared pool, or visible to another country. **The answer:** there is no shared pool — each country runs its own installation, and sharing is technically impossible.



No connection between countries — by design

Could the wrong person get in?

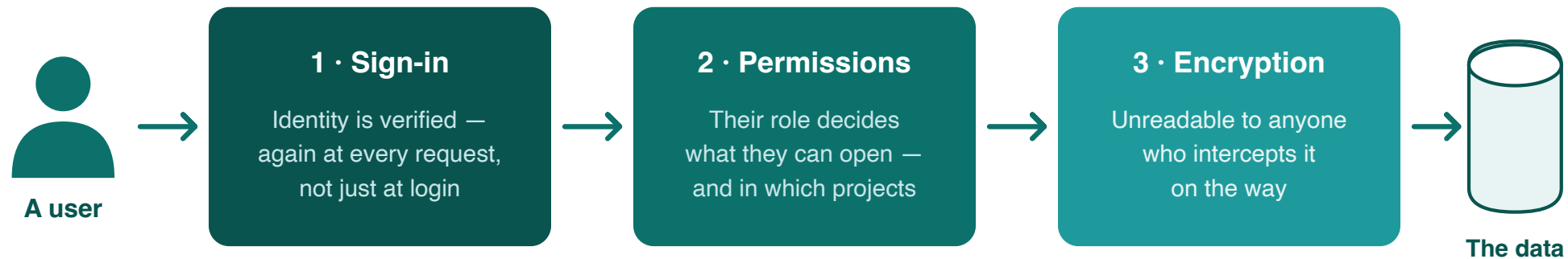
The risk: a leaked password, or an insider reaching beyond their role. **The answer:** accounts, roles, and identity re-checked at every single request.

- The ministry decides **who gets an account**, and each person's role
- Roles set what a person can do — **view, edit, or administer** — and in which projects
- Identity is **re-checked on every request**, not just at login
- Only a **small, named group of administrators** can change the setup

Could data be intercepted — or lost?

The risk: interception on the network, or a server failure taking the data with it. **The answer:** nothing readable leaves the platform, and automatic copies mean everything can be rebuilt.

Three checks stand between a user and the data

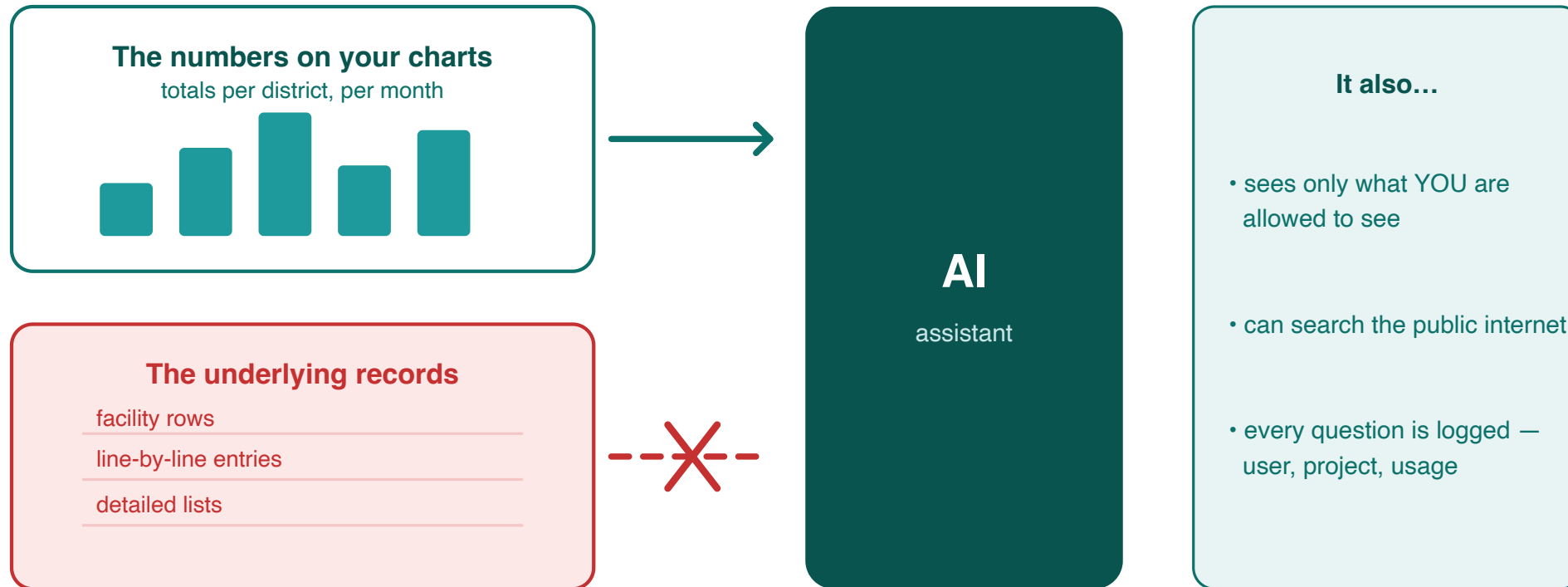


Separately: protection against loss

The database is backed up every 30 minutes, the full installation daily — if a server fails, everything can be rebuilt

Could the AI expose what it sees?

The risk: an assistant that sees too much. **The answer:** it is handed aggregated totals only — the numbers a user could already see, within that user's own permissions. **Never the underlying rows.**



Costs and ownership

The background of the slide is a solid teal color. On the right side, there are several overlapping geometric shapes. These include a large dark teal quarter-circle in the top right, a smaller dark teal quarter-circle in the bottom right, and a series of concentric circles in a lighter teal shade that overlap with the other shapes.

Running costs

Three lines: **server hosting** (fixed), **AI use** (metered), **maintenance** (shared team). No license fees, no per-user fees — the software is open source.



Planning estimates, 2026 prices — no license fees, no per-user charges

The figures — planning estimates

Estimates at 2026 prices and today's usage. **Not a quote.**

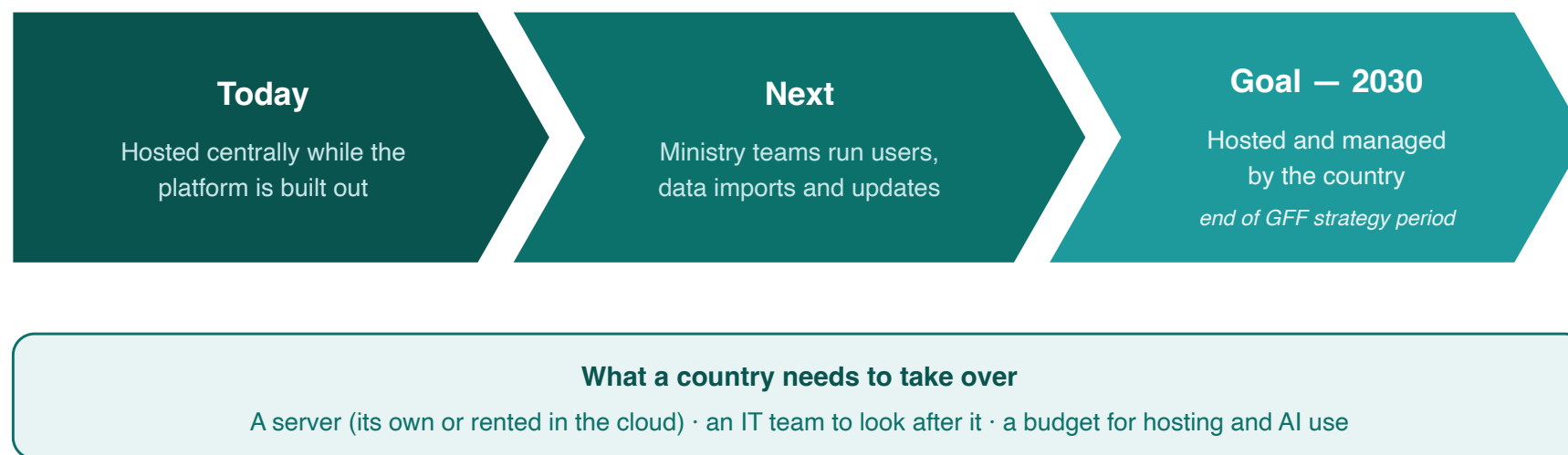
- **Hosting** a country's instance: about **USD 500–800 per year** — larger countries toward the top
- **AI use**: about **USD 350 per month** (~USD 4,000 per year) for a typical country; the largest countries several times this, and it grows with use
- **Shared platform maintenance** (engineering, monitoring, backups): roughly **USD 7,000–10,000 per country per year** today — it falls as more countries join
- **Optional support** (data refresh, quality checks, analysis): about **USD 5,000–15,000 per year**, depending on level
- **Total: about USD 12,000 per year** — up to USD 20,000–30,000 with support
- The lines include the external services: the **login service (Clerk)** and the **AI API (Anthropic)** — no hidden subscriptions
- **Economies of scale are already working**: one team and one codebase serve every country, so the per-country maintenance share falls as more countries join — economics a country gives up by hosting alone

Hosting — and what moving really takes

- Hosted **centrally today**, while the platform is in active development — every country receives fixes and new features the same day
- Built **portable**: the same software runs unchanged on a ministry's own servers — no rebuild needed
- **But migration is a project, not a switch**: readiness assessment, server procurement, team training, a parallel-run period, then cutover — measured in **months, and planned jointly**
- **And hosting alone means carrying alone** what is shared today: maintenance, monitoring, upgrades, and same-day fixes become the ministry team's own workload
- **All of a country's data can be exported in full at any time**, now or later, regardless of hosting

The path to country ownership

For countries who do want to host themselves, there will be effort required from the Ministry of Health. A team will need to run servers, backups, and upgrades; and procurement in place for hosting and for AI services, which are bought from a commercial provider (e.g. Anthropic, OpenAI).



The essentials

- **Monthly totals only** — never patient records
- **One installation per country** — no pathway between countries
- **About USD 12,000 per year** to run — hosting, metered AI, shared maintenance; no license or per-user fees
- **Country hosting by 2030** — the stated transition goal

Technical annex — for IT and DHIS2 teams

Architecture and isolation

- Each country instance is a **separate Docker deployment**: its own application container, its own **PostgreSQL** database, its own **Valkey** cache, its own storage volume, on a private network
- Within an instance, each project has its **own database** (`project_{uuid}`) for authoring content
- Computed results live in **versioned results packages** (parquet files, queried via DuckDB) — projects read the package attached to them, never each other's

Authentication and access control

- Identity is managed by **Clerk** (managed identity provider); session claims are **verified by middleware on every request** — auth bypass exists only for local development and is disabled in production
- **Two-tier RBAC**: instance-level permissions plus per-project roles resolved from the database on each request
- Server access: **SSH by cryptographic key only**, restricted to two named engineers — no password login

Data protection and operations

- **TLS terminates at nginx**, certificates provisioned and auto-renewed via certbot; secrets are injected as environment variables at runtime — never in images, never sent to the browser
- **R analysis modules run in ephemeral containers** (removed on completion) that mount only their own run's sandbox directory
- Backups: **pg_dump every 30 minutes** (3-day rolling window) plus **volume snapshots daily / weekly / monthly**; restores are permission-gated (`can_restore_backups`) and require a server-held key on top of a valid session

AI integration, precisely

- Claude (Anthropic) is reached through a **server-side proxy**; the API key never leaves the server
- Tool calls run **inside the user's authenticated session** — the AI holds no permissions of its own
- Data tools return **aggregated metric outputs only**: there is **no facility-identifier dimension** in the query interface, and any dimension with more than 20 values is summarized as a count
- **Server-side web search/fetch (Anthropic-hosted) is enabled** in the project chat for general questions
- Every request is logged (user, project, model, tokens); **daily per-user and weekly per-instance limits** are enforced

Data and interoperability

- Source data: **aggregated DHIS2 numerators** (facility-month service counts) — imported server-side, with per-(indicator, month) replace semantics and a per-indicator import ledger
- **Full export at any time**: a country's data and results can be exported regardless of hosting arrangement
- The codebase is **open source**: github.com/FASTR-Analytics/platform